

4(a)	Hooke's (law)	B1
4(b)	$k = F/x$ or $k = \text{gradient}$	C1
	= e.g. $12.0 / (0.240 - 0.08)$	A1
	= 75 N m^{-1}	
4(c)	$E = \frac{1}{2}Fx$ or $E = \frac{1}{2}kx^2$ or $E = \text{area under graph}$	C1
	$E = \frac{1}{2} \times 6.0 \times 0.080$ or $\frac{1}{2} \times 75 \times 0.08^2$ = 0.24 J	A1